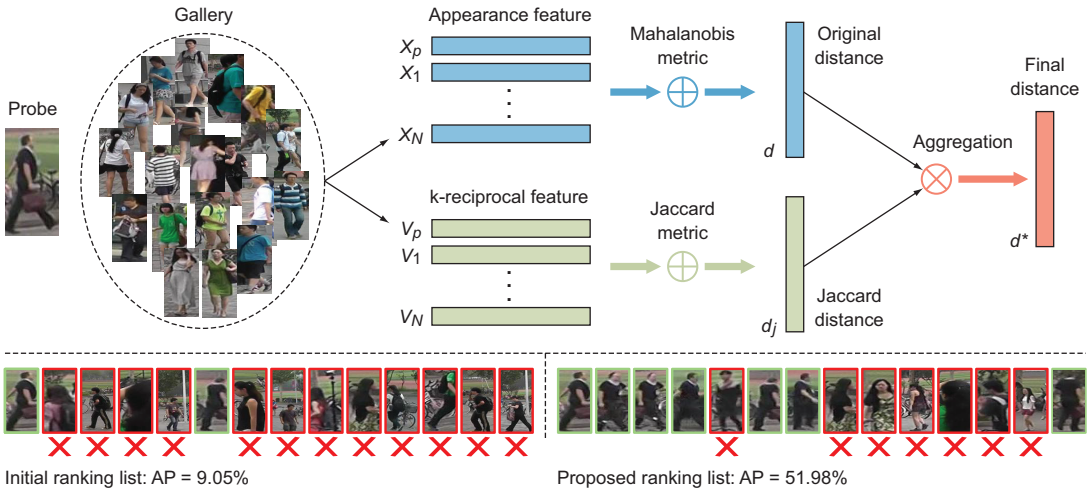

Figure 10.25 Re-ranking proposal by Zhong et al. (*Source:* [28].)

and k-reciprocal feature are extracted for each person. The original distance d and Jaccard distance J_d are calculated for each pair of a probe person and a gallery person. The final distance is then computed as the combination of d and J_d and used to obtain the proposed ranking list.

A recent work in vehicle re-identification, AAVER [25] boosts mAP accuracy by 5% by post-processing using re-ranking.

DEFINITION The *Jaccard distance* is computed among two sets of data and expresses the intersection over the union of the two sets.

- *Tips and tricks*—Luo et al. [29] demonstrated powerful baseline performance on the task of person re-identification. The authors follow the same batch construction from Hermans et al. [11] (studied in this chapter) and use tricks for data augmentation, warm-up learning rate, and label smoothing, to name a few. Noticeably, the authors perform favorably compared to many state-of-the-art methods. You are encouraged to apply these general tricks for training a CNN for any recognition-related tasks.

DEFINITIONS The *warm-up learning rate* refers to a strategy with a learning rate scheduler that modulates the learning rate linearly with respect to a predefined number of initial training epochs. *Label smoothing* modulates the cross-entropy loss so the resulting loss is less overconfident on the training set, thereby helping with model generalization and preventing overfitting. This is particularly useful in small-scale datasets.